AP PRECALCULUS

Unit 1 • Lesson 1.14 practice • Answer key

Lesson 1.14 • Function Model Construction and Application

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

A rectangle has perimeter 40 cm and width w cm. Which function gives its area?

(A) $A(w) = w(40 - w)$

(B) $A(w) = w(20 - w)$

(C) $A(w) = 2w + 40$

(D) $A(w) = w^2 - 20$

Solution

Answer: (B). The length is $\frac{40-2w}{2} = 20 - w$.

Question 2

NO CALCULATOR

What is a reasonable domain for $A(w)$ in the previous question?

(A) $0 < w < 20$

(B) $0 < w < 40$

(C) $w > 0$

(D) All real numbers

Solution

Answer: (A). Both the width w and the length $20 - w$ must be positive.

Question 3

NO CALCULATOR

What is the maximum area of a rectangle with perimeter 40 cm?

(A) 400 cm^2

(B) 200 cm^2

(C) 100 cm^2

(D) 80 cm^2

Solution

Answer: (C). $A(w) = -(w - 10)^2 + 100$, maximum 100 at $w = 10$.

Question 4

CALCULATOR

An open box is made from a 12 in by 12 in square sheet by cutting squares of side x from the corners. What is the maximum volume?

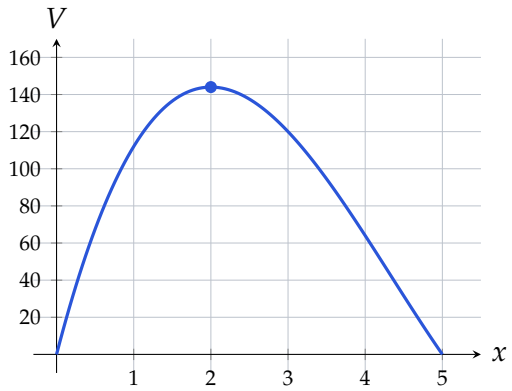
(A) 96 in^3
(B) 108 in^3
(C) 144 in^3
(D) 128 in^3

Solution
Answer: (D). $V(x) = x(12-2x)^2$ on $0 < x < 6$. The maximum is at $x = 2$: $V(2) = 2 \cdot 8^2 = 128$ in^3.

Question 5 (graph)

NO CALCULATOR

The graph shows the volume $V(x)$, in cm^3, of an open box made from a 10 cm by 16 cm sheet with squares of side x cm cut from the corners. Which value of x gives the greatest volume?



(A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 5$

Solution
Answer: (B). The highest point of the graph is at $x = 2$, where $V(2) = 2 \cdot 6 \cdot 12 = 144$ cm^3.

Question 6 (Free response)

CALCULATOR

A cylindrical can must hold 500 cm^3. Let r be its radius and h its height, in centimeters.

(a) Express h in terms of r.
(b) Show that the surface area is $S(r) = 2\pi r^2 + \dfrac{1000}{r}$.
(c) Use a graphing calculator to find the radius that minimizes S, and the minimum surface area.
(d) Describe the behavior of S as $r \to 0^+$ and as $r \to \infty$, and explain why in context.

Solution
(a) $\pi r^2 h = 500$, so $h = \dfrac{500}{\pi r^2}$.
(b) $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \cdot \dfrac{500}{\pi r^2} = 2\pi r^2 + \dfrac{1000}{r}$.
(c) The minimum is at $r \approx 4.30$ cm, with $S \approx 348.7$ cm^2 (then $h \approx 8.60$ cm).
(d) $S \to \infty$ in both cases. A very thin can must be very tall (the $\frac{1000}{r}$ term grows); a very wide can has huge top and bottom (the $2\pi r^2$ term grows).